

Data Science Aptitude Test — Set 2

More Important Questions & Answers

Quantitative Aptitude | Logical Reasoning | Data Interpretation | Statistics & ML
Fundamentals | SQL & Python Aptitude

A fresh, non-overlapping set of questions to complement Set 1

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This set covers different questions from Set 1 so you can use both for broader practice. Try solving each on paper before checking the answer.

1. Quantitative Aptitude

Speed-focused arithmetic and problem solving, commonly used to screen candidates before the technical round.

Q1. A container has milk and water in the ratio 5:3. If 16 litres of the mixture is water, how much milk does the container have?

Answer: Ratio parts: 5 (milk) + 3 (water) = 8 parts. 3 parts = 16 litres $\rightarrow$ 1 part = $16/3$ litres. Milk = 5 parts = $5 \times 16/3 = 80/3 \approx 26.67$ litres.

Tip: Always convert a ratio to 'per part' value first, then scale up to the required quantity.

Q2. A alone can finish a job in 15 days, and with the help of B, they finish it in 6 days. In how many days can B alone finish the job?

Answer: A's rate = $1/15$. Combined rate = $1/6$. B's rate = $1/6 - 1/15 = 5/30 - 2/30 = 3/30 = 1/10$. So B alone takes 10 days.

Tip: Subtract rates (not times) when isolating one worker's contribution from a combined rate.

Q3. The compound interest on ₹20,000 for 2 years at 10% per annum is how much more than the simple interest for the same period and rate?

Answer: SI = $(20000 \times 10 \times 2)/100 = ₹4,000$. CI = $20000 \times (1.1)^2 - 20000 = 20000 \times 1.21 - 20000 = 24200 - 20000 = ₹4,200$. Difference = $4200 - 4000 = ₹200$.

Tip: For 2 years, the CI-SI difference simplifies to $P \times (R/100)^2$ — here $20000 \times 0.01 = 200$, a useful shortcut.

Q4. A car covers a certain distance at 60 km/hr and returns over the same distance at 40 km/hr. Find the average speed for the whole journey.

Answer: For equal distances, average speed = $(2 \times v_1 \times v_2)/(v_1 + v_2) = (2 \times 60 \times 40)/(60 + 40) = 4800/100 = 48$ km/hr.

Tip: Never simply average the two speeds (which would wrongly give 50) when distances are equal, not times.

Q5. How many ways can a committee of 3 people be selected from a group of 7 people?

Answer: This is a combination problem: $C(7,3) = 7! / (3! \times 4!) = (7 \times 6 \times 5)/(3 \times 2 \times 1) = 210/6 = 35$.

Tip: Use combinations (nCr) whenever the order of selection doesn't matter, unlike permutations.

Q6. The price of an item is first increased by 20% and then decreased by 20%. What is the net percentage change in price?

Answer: Let price = 100. After +20%: 120. After -20% of 120: $120 - 24 = 96$. Net change = $96 - 100 = -4\%$, i.e. a 4% decrease.

Tip: Successive % increase then equal % decrease never cancels out — it always results in a net decrease.

Q7. Two dice are rolled together. What is the probability that the sum of the numbers shown is 8?

Answer: Total outcomes = $6 \times 6 = 36$. Pairs summing to 8: (2,6),(3,5),(4,4),(5,3),(6,2) = 5 outcomes. Probability = $5/36$.

Tip: For two-dice sum problems, list ordered pairs carefully — (2,6) and (6,2) are counted separately.

2. Logical & Analytical Reasoning

Pattern spotting and structured deduction, testing the same 'break the problem down' skill used in data analysis.

Q8. Find the missing number in the series: 3, 7, 15, 31, 63, ?

Answer: Each term follows the pattern (previous $\times 2$) + 1: $3 \times 2 + 1 = 7$, $7 \times 2 + 1 = 15$, $15 \times 2 + 1 = 31$, $31 \times 2 + 1 = 63$, so next = $63 \times 2 + 1 = 127$.

Tip: When ratios aren't constant, test 'double and add/subtract a constant' as the next pattern.

Q9. In a certain code, 'MOUSE' is written as 'NPVTF'. How is 'TIGER' written in that code?

Answer: Each letter is shifted forward by 1 (M $\rightarrow$ N, O $\rightarrow$ P, U $\rightarrow$ V, S $\rightarrow$ T, E $\rightarrow$ F). Applying the same shift: T $\rightarrow$ U, I $\rightarrow$ J, G $\rightarrow$ H, E $\rightarrow$ F, R $\rightarrow$ S = UJHFS.

Tip: Confirm the shift direction and amount using every letter in the example word, not just the first.

Q10. Statements: All engineers are graduates. No graduate is uneducated. Conclusion: No engineer is uneducated. Is the conclusion valid?

Answer: Valid. Since all engineers fall within the graduate set, and no graduate is uneducated, engineers must also fall outside the 'uneducated' set — the conclusion follows logically.

Tip: When the middle term (graduates) fully links the two statements, the conclusion is usually valid.

Q11. Introducing a woman, a man said, 'Her father is the only son of my father.' How is the woman related to the man?

Answer: 'The only son of my father' refers to the man himself. So the woman's father is the man — meaning the woman is the man's daughter.

Tip: Whenever a puzzle says 'only son/daughter of my father/mother', it's referring back to the speaker.

Q12. Arrange the following in a logical order: 1. Data Cleaning, 2. Model Deployment, 3. Data Collection, 4. Model Training, 5. Exploratory Data Analysis.

Answer: Correct order: 3 (Data Collection) $\rightarrow$ 1 (Data Cleaning) $\rightarrow$ 5 (Exploratory Data Analysis) $\rightarrow$ 4 (Model Training) $\rightarrow$ 2 (Model Deployment).

Tip: Sequencing questions in DS aptitude tests often test whether you understand the real project workflow.

Q13. Choose the odd one out: Mean, Median, Mode, Variance.

Answer: Variance is the odd one out. Mean, Median, and Mode are all measures of central tendency, while Variance is a measure of spread (dispersion), not central tendency.

Tip: Classification questions like this test whether you know which 'family' a statistical term belongs to.

3. Data Interpretation

Reasoning over numeric summaries — a practical preview of everyday data-scientist tasks.

Q14. A dataset has quartiles $Q1 = 20$ and $Q3 = 50$. What is the Interquartile Range (IQR), and what is the upper bound for detecting outliers using the standard $1.5 \times \text{IQR}$ rule?

Answer: $\text{IQR} = Q3 - Q1 = 50 - 20 = 30$. Upper bound = $Q3 + 1.5 \times \text{IQR} = 50 + 1.5 \times 30 = 50 + 45 = 95$. Any value above 95 would be flagged as a potential outlier.

Tip: Remember both bounds: Lower = $Q1 - 1.5 \times \text{IQR}$, Upper = $Q3 + 1.5 \times \text{IQR}$.

Q15. A survey of 200 people shows 120 prefer Python and 90 prefer R, with 30 people preferring both. How many people prefer neither?

Answer: Using inclusion-exclusion: People who prefer at least one = $120 + 90 - 30 = 180$. People who prefer neither = $200 - 180 = 20$.

Tip: Whenever 'both' is mentioned, use $\text{Total} = A + B - \text{Both} + \text{Neither}$ to avoid double-counting.

Q16. A company's user base grew from 2,000 to 4,500 over 3 years. Find the approximate Compound Annual Growth Rate (CAGR).

Answer: $\text{CAGR} = (\text{End}/\text{Start})^{(1/n)} - 1 = (4500/2000)^{(1/3)} - 1 = (2.25)^{(1/3)} - 1 \approx 1.31 - 1 = 0.31$, i.e. approximately 31% per year.

Tip: CAGR uses a geometric (multiplicative) average, not a simple arithmetic average of yearly growth rates.

Q17. A confusion matrix for a binary classifier shows: True Positives = 40, False Positives = 10, False Negatives = 5, True Negatives = 45. Calculate the model's accuracy.

Answer: $\text{Accuracy} = (\text{TP} + \text{TN}) / \text{Total} = (40 + 45) / (40 + 10 + 5 + 45) = 85/100 = 0.85$, i.e. 85%.

Tip: Total predictions = $\text{TP} + \text{FP} + \text{FN} + \text{TN}$ — a common mistake is forgetting to include all four cells.

4. Statistics & Machine Learning Fundamentals

Concept-level questions that test core statistical and ML intuition expected from a data science candidate.

Q18. Using the confusion matrix from the previous section (TP=40, FP=10, FN=5, TN=45), calculate Precision and Recall.

Answer: Precision = $TP/(TP+FP) = 40/(40+10) = 40/50 = 0.80$ (80%). Recall = $TP/(TP+FN) = 40/(40+5) = 40/45 \approx 0.889$ (88.9%).

Tip: Precision answers 'of predicted positives, how many were correct?'; Recall answers 'of actual positives, how many did we catch?'

Q19. What is overfitting in machine learning, and name one common technique to reduce it.

Answer: Overfitting occurs when a model learns the training data too closely — including its noise — so it performs well on training data but poorly on unseen/test data. Common fixes include regularization (L1/L2), cross-validation, pruning, dropout (for neural networks), or simply gathering more training data.

Tip: Mention the training-vs-test accuracy gap as the practical symptom interviewers look for.

Q20. What is the Central Limit Theorem, and why is it important in statistics?

Answer: The Central Limit Theorem states that the sampling distribution of the sample mean approaches a normal distribution as the sample size grows large, regardless of the population's original distribution. It is important because it justifies using normal-distribution-based methods (like confidence intervals and hypothesis tests) even when the underlying data isn't normal.

Tip: A common follow-up: state the typical rule of thumb that $n \geq 30$ is 'large enough' for CLT to apply.

Q21. What is multicollinearity in regression, and why is it a problem?

Answer: Multicollinearity occurs when two or more independent variables in a regression model are highly correlated with each other. It's a problem because it makes it difficult to isolate the individual effect of each variable, inflates the standard errors of coefficients, and can make coefficient estimates unstable.

Tip: Mention VIF (Variance Inflation Factor) as the standard metric used to detect multicollinearity.

Q22. What is the difference between supervised and unsupervised learning? Give one example of each.

Answer: Supervised learning uses labeled data (input-output pairs) to train a model, e.g. predicting house prices from features (regression). Unsupervised learning works with unlabeled data to find hidden patterns or structure, e.g. grouping customers into segments using clustering (like K-Means).

Tip: If asked for a third category, mention reinforcement learning (learning via rewards/penalties).

Q23. What does it mean for an estimator to be 'unbiased', in simple terms?

Answer: An estimator is unbiased if its expected value equals the true parameter it is estimating, across many repeated samples — meaning it doesn't systematically overestimate or underestimate the true value on average.

Tip: This is exactly why sample variance divides by $(n-1)$ instead of n — to remove that systematic bias.

Q24. What is the difference between a Type I error and the significance level (alpha) in hypothesis testing?

Answer: The significance level (alpha) is the threshold probability the researcher sets in advance (commonly 0.05) for rejecting the null hypothesis. A Type I error is the actual event of wrongly rejecting a true null hypothesis. In other words, alpha is the pre-set risk tolerance, while Type I error is what happens when that risk materializes.

Tip: Interviewers use this to check you understand alpha as a decision rule, not just a definition.

Q25. What is the purpose of train-test split (or train-validation-test split) in a machine learning workflow?

Answer: It separates data so the model is trained on one portion and evaluated on data it has never seen, giving an honest estimate of how well the model will generalize to new, real-world data. A validation set is often added separately from the test set to tune hyperparameters without 'leaking' information from the final test set.

Tip: Mention that the test set should only be used once, at the very end, to avoid overfitting to it indirectly.

5. SQL & Python Aptitude

Practical technical-aptitude questions frequently paired with quant/reasoning rounds for data roles.

Q26. What is the difference between DELETE, TRUNCATE, and DROP in SQL?

Answer: DELETE removes specific rows based on a WHERE condition and can be rolled back; it's slower since it logs each row. TRUNCATE removes all rows from a table quickly but keeps the table structure. DROP removes the entire table (structure and data) from the database permanently.

Tip: Ranking by 'how much they remove': DELETE (rows) < TRUNCATE (all rows) < DROP (whole table).

Q27. Write a SQL query to find duplicate email entries in a table called Users(id, email).

Answer: SELECT email, COUNT(*) FROM Users GROUP BY email HAVING COUNT(*) > 1; This groups rows by email and keeps only groups that appear more than once.

Tip: This GROUP BY + HAVING combination is one of the most frequently asked SQL aptitude patterns.

Q28. What is a Primary Key, and how is it different from a Foreign Key?

Answer: A Primary Key uniquely identifies each row in its own table and cannot contain NULL values. A Foreign Key is a column (or set of columns) in one table that references the Primary Key of another table, used to enforce a relationship between the two tables.

Tip: Think of Primary Key as 'my ID' and Foreign Key as 'a pointer to someone else's ID'.

Q29. In Python, what will be the output of: sorted([5, 2, 9, 1], reverse=True)?

Answer: The output is [9, 5, 2, 1]. The sorted() function returns a new sorted list, and reverse=True sorts it in descending order instead of the default ascending order.

Tip: Remember sorted() returns a new list, while list.sort() modifies the list in place and returns None.

Q30. What is the difference between a Python dictionary and a list, and when would you use each?

Answer: A list is an ordered collection accessed by numeric index, useful for sequences of similar items. A dictionary stores key-value pairs and is accessed by key, useful when you need fast lookups by a meaningful label (like mapping a student's name to their marks).

Tip: Dictionaries use hashing internally, giving average O(1) lookup time versus O(n) for searching a list.

Q31. What does a SQL window function like RANK() OVER (PARTITION BY ... ORDER BY ...) do, and how is it different from GROUP BY?

Answer: A window function computes a value (like a rank) across a set of rows related to the current row, without collapsing those rows into a single output row — unlike GROUP BY, which reduces each group into one summary row. PARTITION BY defines the groups, and ORDER BY defines the ranking order within each group.

Tip: Use window functions when you need per-row detail alongside a group-level calculation, e.g. ranking employees by salary within each department while still listing every employee.

How to Use This Set

- Use this set alongside Set 1 for wider topic coverage — together they span quant, reasoning, DI, stats/ML, and SQL/Python.
- The Statistics & ML Fundamentals section is the highest-value section for a data science role — revise it most frequently.
- Practice the SQL queries in an actual database tool rather than only reading them, to build muscle memory.
- Re-attempt the Data Interpretation questions with different numbers to make sure you understand the method, not just this specific answer.

All the best for your data science aptitude test!